Week 7: SQL Query Implementation for E-Commerce Database

Scenario

The e-commerce company has collected customer, product, order, and inventory data. The management team wants to retrieve useful information from the database for daily operations and basic business analysis.

Students will apply SQL queries to search, filter, sort, and generate reports from the existing E-Commerce Database.

1. Perform Basic SQL Queries

Concept

SQL SELECT command is used to retrieve data from database tables.

1. Display all customer details.

```
SQL> SELECT * FROM CUSTOMER;

                          CUSTOMER_ID CUSTOMER_NAME        EMAIL
------------------------------------ -------------------- ---------------
                                 101 Ananya               ananya@gmail.co
                                 102 Priya                priya@gmail.com
                                 103 Kaviya               kaviya@gmail.co
                                 104 Karu                 karu@gamil.com
                                 105 Gobi                 gobi@gmail.com
```

2. Display all available products.

```
SQL> SELECT * FROM PRODUCT;

PRODUCT PRODUCT_NAME         CATEGOR     PRICE STOCK_QUANTITY
------- -------------------- ------- --------- --------------
11      Red Rose             1           50.00            100
12       White Lily          2           80.00             60
13       Jasmine             3          550.00            150
14      Purple Orchid        4          120.00             50
15       Rose Bouquet        5          350.00             30
```

3. Retrieve product names and prices only.

```
SQL> SELECT PRODUCT_NAME, PRICE FROM PRODUCT;

PRODUCT_NAME              PRICE
------------------- ---------
Red Rose                  50.00
 White Lily               80.00
 Jasmine                 550.00
Purple Orchid            120.00
 Rose Bouquet            350.00
```

4. Display all orders placed by customers.

```
SQL> SELECT * FROM Orders;

ORDER_ID                                CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT OR
---------- ---------------------------------------- --------- ------------ --
0001                                            101 01-SEP-26         3200 Pe
0002                                            102 03-SEP-26         1299 Co
0003                                            103 05-SEP-26         1000 Re
0004                                            104 23-DEC-26         1500 De
0005                                            105 31-JUL-08          150 Re
```

5. Retrieve payment details.

```
SQL> SELECT * FROM Payment;

PAYMENT ORDER_I PAYMENT_D PAYMENT_MODE           PAYMENT PAYMENT_STATUS
------- ------- --------- -------------------- ---------- ------------------
PAY101  0001    03-SEP-26 UPI                             COMPLETED
PAY102  0002    04-SEP-26 CARD                            FAILED
PAY103  0003    09-SEP-26 NET BANKING                     PENDING
PAY104  0004    25-DEC-26 CARD                            FAILED
PAY105  0005    14-AUG-08 CASH                            COMPLETED
```

2. Apply Filtering Conditions Using WHERE

6. Find products with price greater than ₹500.

```
SQL> SELECT * FROM Product WHERE Price > 500;

PRODUCT PRODUCT_NAME         CATEGOR     PRICE STOCK_QUANTITY
------- -------------------- ------- --------- --------------
13       Jasmine             3          550.00            150
```

7.

8. Display products available in stock.

```
SQL> SELECT * FROM Inventory WHERE Stock_Quantity > 0;

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
------- ------- ------- -------------- ----------- ---------
201     11      S10                100  Available  20-SEP-26
202     12      S20                 60 Available   21-SEP-26
203     13      S30                150 Available   22-SEP-26
204     14      S40                 25 Low Stock   23-SEP-26
205     15      S50                105  Available  24-SEP-26
```

9. Display Delivered orders.

```
SQL> SELECT * FROM Orders WHERE Order_Status = 'Delivered';

ORDER_ID                                 CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT OR
---------- ---------------------------------------- --------- ------------ --
0004                                            104 23-DEC-26         1500 De
```

10. Find products with rating above 4.

```
SQL> SELECT * FROM Product WHERE Rating > 4;

PRODUCT_ID   PRODUCT_NAME         CATEGOR     PRICE STOCK_QUANTITY RATING
---------- -------------------- ------- --------- -------------- ------
11           Red Rose             1          50.00            100    4.5
13            Jasmine             3         550.00            150    4.8
14           Purple Orchid        4         120.00             50    4.2
```

3. Sort Data Using ORDER BY

Tasks

11. Display products from lowest to highest price.

```
SQL> SELECT * FROM Product ORDER BY Price ASC;

PRODUCT_ID   PRODUCT_NAME         CATEGOR     PRICE STOCK_QUANTITY RATING
---------- -------------------- ------- --------- -------------- ------
11           Red Rose             1          50.00            100    4.5
12            White Lily          2          80.00             60    3.5
14           Purple Orchid        4         120.00             50    4.2
15            Rose Bouquet        5         350.00             30    3.8
13            Jasmine             3         550.00            150    4.8
```

12. Display customers alphabetically.

```
SQL> SELECT CUSTOMER_NAME FROM CUSTOMER
  2  ORDER BY CUSTOMER_NAME ASC;

CUSTOMER_NAME
--------------------
Ananya
Gobi
Karu
Kaviya
Priya
```

13. Find top expensive products.

```
SQL> SELECT * FROM Product ORDER BY Price DESC;
truncating (as requested) before column RATING


PRODUCT PRODUCT_NAME
------- ------------------------------------------------------------
-- ------- ---------- -------------
13       Jasmine
15       Rose Bouquet
14      Purple Orchid
12       White Lily
11      Red Rose
```

14. Display latest orders first.

```
SQL> SELECT * FROM ORDERS
  2  ORDER BY ORDER_DATE DESC;

ORDER_ID   CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
---------- ----------- --------- ------------ --------------
0004               104 23-DEC-26         1500 Delivered
0003               103 05-SEP-26         1000 Ready
0002               102 03-SEP-26         1299 Confirmed
0001               101 01-SEP-26         3200 Pending
0005               105 31-JUL-08          150 Ready
```

4. Retrieve Unique Values Using DISTINCT

Tasks

15. Display unique product categories.

```
SQL> SELECT * FROM Product ORDER BY Price ASC;

PRODUCT_ID   PRODUCT_NAME         CATEGOR     PRICE STOCK_QUANTITY RATING
---------- -------------------- ------- --------- -------------- ------
11           Red Rose             1           50.00            100    4.5
12            White Lily          2           80.00             60    3.5
14           Purple Orchid        4          120.00             50    4.2
15            Rose Bouquet        5          350.00             30    3.8
13            Jasmine             3          550.00            150    4.8
```

16. Find different payment methods used by customers.

17.
```
SQL> SELECT c.customer_name, o.order_id, o.order_date,o.order_status FROM
stomer c JOIN orders o ON  c.customer_id = o.customer_id;

CUSTOMER_NAME   | ORDER_ID   | ORDER_DAT | ORDER_STATUS
--------------- | ---------- | --------- | --------------------
Ananya          | 0001       | 01-SEP-26 | Pending
Priya           | 0002       | 03-SEP-26 | Confirmed
Kaviya          | 0003       | 05-SEP-26 | Ready
Karu            | 0004       | 23-DEC-26 | Delivered
Gobi            | 0005       | 31-JUL-08 | Ready
```

5. Search Products Based on Conditions

Tasks

Write SQL queries to:

18. Find products within a specific price range. (Product Between 20,000 to 40,000)
19. Display products belonging to a particular category.
20. Find products currently available in inventory.
21. Search products using product name.
22. Find low-stock products.

```
SQL> SELECT * FROM Inventory WHERE Stock_Quantity > 0;

INVENTO PRODUCT SELLER_ STOCK_QUANTITY STOCK_STATU LAST_UPDA
------- ------- ------- -------------- ----------- ---------
201     11      S10                100  Available  20-SEP-26
202     12      S20                 60 Available   21-SEP-26
203     13      S30                150 Available   22-SEP-26
204     14      S40                 25 Low Stock   23-SEP-26
205     15      S50                105  Available  24-SEP-26
```

6. Retrieve Customer and Product Information

Tasks

23. Display customer details with their orders.
24. Display product details with category information.
25. Find customers who purchased a specific product.
26. Display products purchased by each customer.

7. Apply Multiple Filtering Conditions

Tasks

27. Find electronics products costing more than ₹10000.
28. Search products containing the word "Mobile".
29. Find orders placed within a specific date range.

8. Generate Basic Business Reports

Report 1: Product Availability Report

Display:

- Product name
- Price
- Stock quantity
- Availability status

Report 2: Customer Report

Generate:

- Total number of customers
- New customer registrations

Report 3: Order Report

Display:

- Total orders
- Completed orders
- Pending orders
- Cancelled orders

Report 4: Product Performance Report

Find:

- Highest-priced products
- Most reviewed products
- Available products